

PARAM KAMLESHKUMAR PATEL

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EDUCATION

Stony Brook University

MS in Computer Science (GPA: 3.70/4.00)

August 2024 – May 2026

Stony Brook, NY, USA

Indus University

B.Tech in Computer Engineering (GPA: 9.90/10.00)

August 2020 – May 2024

Ahmedabad, GJ, India

TECHNICAL SKILLS

Languages: Java, Go, Python, C++, SQL, JavaScript, TypeScript

Frameworks: Spring Boot, FastAPI, Node.js/Express, Next.js, React, Angular

AI/ML: Generative AI, LLMs, RAG, Multi-Agent Systems, LangChain, Prompt Engineering, Vector Search, OpenAI API

Distributed Systems: gRPC, Microservices, REST, Consensus (Paxos/RAFT/PBFT), Concurrency Control

Database: PostgreSQL, MySQL, MongoDB, Redis, pgvector

Cloud / DevOps: Docker, Kubernetes, Linux, AWS (EC2/S3/Lambda), GitHub Actions, CI/CD

Tools: Git, Postman, PyTest, JUnit, Splunk, Dynatrace

EXPERIENCE

Arch MI

June 2025 – August 2025

Generative AI Intern

Greensboro, NC, USA

- Engineered a multi-agent, self-healing IT automation platform integrating specialized agents for tools such as Splunk, Dynatrace, Veracode, GitHub, and Outlook, featuring a modular architecture for unlimited future integrations.
- Built a supervisor agent to orchestrate user requests and aggregate outputs from specialized agents, enabling a full self-healing workflow and powering an Angular-based chatbot interface for real-time system insights.
- Streamlined automated GitHub PR generation and CI/CD pipelines for validated fixes, reducing incident resolution time by 75% and saving 3+ hours per issue.

Uminber Designs

January 2024 – April 2024

AI Intern

Ahmedabad, GJ, India

- Deployed a Retrieval-Augmented Generation (RAG) chatbot leveraging OpenAI and LangChain for enhanced document Q&A, driving a 30% increase in user adoption.
- Applied prompt tuning and modular chain refinement through LangChain and LangSmith, improving accuracy by 25%.
- Launched a full-stack RAG application using Next.js, PostgreSQL, and Docker in production, achieving 99% uptime.

Itorix Infotech LLP

July 2023 – August 2023

Web Development Intern

Pune, MH, India

- Constructed Java-based REST APIs and core back-end logic for a real-time RFID tracking system, increasing data processing speed by 30% through optimized object-oriented design.
- Architected 10+ relational SQL tables and developed complex stored procedures, reducing data retrieval latency by 40% for high-volume asset monitoring reports.
- Programmed Python back-end solutions for cross-functional teams, automating manual data validation and saving department stakeholders 15+ hours per week.

PROJECTS

Fault-tolerant Distributed Transaction System

Param2k23/Distributed-Transaction-System

- Developed a concurrency-friendly Golang/gRPC-based distributed transaction system for a banking use case, achieving 99.9% durability and sub-500ms latency via robust data structures and advanced concurrency control.
- Implemented Multi-Paxos, optimized RAFT, and PBFT for consensus, incorporating leader election, log replication, checkpointing, and efficient message serialization across 20+ replicas.
- Engineered a scalable key-value store with a custom RAFT consensus layer, ensuring stable performance in asynchronous multi-threaded environments with minimal overhead for high-throughput CRUD operations.

Multi-Agent AI Platform

Param2k23/Enterprise-AI-Platform

- Designed and deployed a multi-agent AI platform with 6 specialized agents using LangChain, semantic search, and autonomous task orchestration, successfully executing 1,500+ multi-step workflows across research and reasoning tasks.
- Built a Retrieval-Augmented Generation (RAG) pipeline using PostgreSQL, pgvector, embeddings, and Redis caching, indexing 50,000+ documents while maintaining sub-250ms retrieval latency for semantic search.
- Implemented Dockerized FastAPI microservices supporting 500+ concurrent requests for scalable agent communication.

ACHIEVEMENTS

3x Hackathon Winner — HackNYU, Hopper Hacks, Arch MI Hackathon